Realistic simulation of a cell membrane made with phospholipids

Objectives: Create a realistic model of a cell membrane, composed by DMPC phospholipids in an aqueous 0.15 M sodium-chloride solution, then run a full MD simulation (equilibration + production) and analyze microscopical structure of several atomic species.

Tools: NAMD (MD simulation) and VMD (visualization and analysis). To install them, go to University of Illinois-Urbana Champaign:
https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD
https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=VMD

STEP 1: Preparing Setup. Once VMD has been installed, visualize your system, generated previously (download the set of files 'charmm-gui-DMPC-NaCl-membrane.tgz' from "ATENEA" and unzip it.

Use the instructions:

1. 'vmd ';
2. File -> New molecule -> Browse: select 'step5_assembly.psf '(+ 'Open') -> Load;
and then do
3. Browse: 'step5_assembly.pdb (+ 'Open') -> Load...and have a look.
HINT: Try to use the tools "Graphics -> Representations -> etc. You can use the VMD user guide: https://www.ks.uiuc.edu/Research/vmd/current/ug/

(Next steps after NAMD has been installed)

STEP 2: Equilibrating the system.

Series of jobs:

1. Go to directory > charmm-gui-DMPC+NaCl+water/namd
2. Copy the downloaded NAMD files here (namd2, charmrun, parameters, psfgen, etc.)
3. Make "./namd2 step6.1_equilibration.inp"

This will run 25000 steps' energy minimization, it may take 2-3 hours, depending on architecture and processor of your laptop.

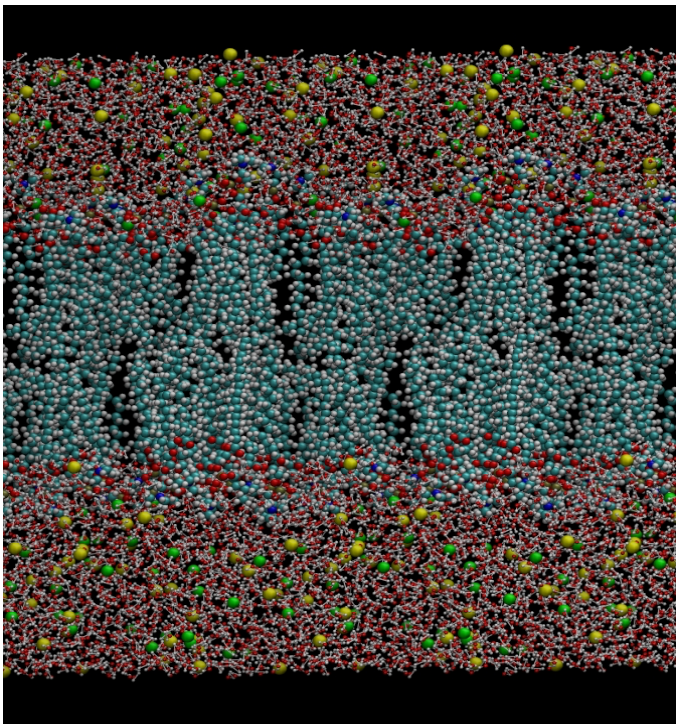
4. After first run of the series, the remaining equilibration steps must be taken, so make "./namd2 step6.N_equilibration.inp" for N=2-6, 25000 or 50000 steps, it may take several hours.

STEP 3: Production run + analysis of the data collected.

Now you should make the production run: "./namd2 step7_production.inp > output.log &" (500000 steps, total length 0.25 ns, if time step is set to 0.5 fs). Several hours...or days. Estimate how long will it take before running it.
After the long run ends, have a look at the final movie. To visualize it, use VMD:

/home/.../charmm-gui/vmd step5_assembly.psf ./namd/step7_production.dcd &

Your system should look like this:



1. Check the energy, temperature and pressure versus time. With the "log" file "output.log", you can use the tool "NAMD plot". You should get energy and temperature fluctuating around given mean values. What happens to pressure? Explain the results. HINT: Plot 'PRESSURE' and PRESSURE-AVERAGE' together and also check reference "F.Jähnig, Biophys. J. 71, 1348 (1996)."

2. STRUCTURE: Radial distribution functions of Ow-Hw, Ow-Na, Hw-Cl, N-Na, N-Cl, P-Na, P-Cl: Use 'Extensions' -> 'Analysis' -> 'Radial Pair Distribution Function g(r)'. In 'Selection 1' and '2', write: 'type OT' and 'type HT' for g(r)_{OH}. The numbers of frames run from 0 to N-1 (according the count shown at the main screen). Use 'type CLA' for chloride, 'type SOD' for sodium, 'type NTL' for nitrogen and 'type PL' for phosphorous. Obtain the corresponding coordination numbers n(r) to get complementary information: use 'Display Int(g(r))' from the same 'Radial' tool. Explain your results in physical terms, using the atomic electrical forces as “leitmotif”.

3. Hydrogen bonding: Using the tool Extensions' -> 'Analysis' -> 'Hydrogen Bonds', calculate the number of HB per water molecule in the system. Set 'Selection 1' as 'resname TIP3' and set HB definition to 'Donor-Acceptor distance = 3.6 Å' and 'Angle cutoff = 30'. Knowing that bulk water has about 4 HBs per molecule, explain your results.

FINAL REMARKS:

- Play with 'Graphics' then 'Representations' to customize your view.
- You can save a snapshot of your system at 'File' then 'Render'.
- And you can save a movie of your simulation with tool 'Movie Maker' from 'Extensions' then 'Visualization'.
- A recent reference of similar simulations is by Jurkiewicz et al. Biochimica et Biophysica Acta 1818 (2012) 609–616 (uploaded in ATENEA)

QUESTIONS:

1. Based on what you learned up to now, explore the influence of temperature on microscopic properties of water, ion and lipid. Hint: Try equilibration + “short” production runs (about 300000 steps) at 283 and 323 K.

2. What is the effect of the size on your simulations? Try N = 50, 100, 150 lipids. To do this, you should use the tool CHARMM-GUI (http://www.charmm-gui.org/?doc=input/membrane)

3. How could you make the simulations more realistic? What particles could be included? Try a case (for instance N = 50, T = 303 K) including such particles. To do this, prepare a setup with some lipids replaced by those additional particles.